

**LISTENING**
#28**Teacher's Guide****SCRIPT: GROWING IN SPACE**

Presenter: The Japanese astronaut Norishige Kanai has been on board the International Space Station for the past three weeks. And today he sent out this eye catching tweet:

"Good morning everybody. I have a major announcement today. We had our bodies measured after reaching space and wow wow wow. I had actually grown by as much as nine centimeters."

Mr. Kanai I went on to say that he was a bit worried that he wouldn't fit in the seat allocated for him in the Soyuz spacecraft, which will take him back to Earth. Chris Hadfield is a Canadian former astronaut. He spoke to the BBC's Tim Franks.

Hadfield: It's more than average for sure. And of course it's not really growth. You're not growing again like when you were a teenager. What's actually happening is each one of the bones in your body is getting slightly further apart because there's no gravity to push you down anymore. And so especially up your backbone, each of the little vertebra gets a little further apart. And if you're a tall astronaut like Norishige, then it gets magnified because you're longer to begin with. But nine centimeters, that's probably more than reality. It's really hard to measure somebody when there's no gravity. You can't just stand up against a wall. It's kind of like measuring yourself floating in the surf. So it will be interesting to see when they remeasure exactly how much taller he's gotten.

Franks: OK. It's one of those problems about being in space that I hadn't really come up with, yes, that it is difficult to get a tape measure up against somebody. Did it happen to you when you were aboard the space station?

Hadfield: It does. All astronauts, their backbones stretch. As you get a little taller when you're weightless, and in my case since I'm about six feet tall, I got about maybe four centimeters taller. You know about that much - a little bit bigger. And what's intriguing is that you have to anticipate that. What Norishige was talking about is the seat that will support and protect his body when he comes crashing back to Earth on the Soyuz. It's a crash seat. It's designed to take up the force of a landing. And so you don't want it to fit improperly. And the Russian designers, the sculptors, who build those seats actually anticipate us being a little bit taller. In his case, you want to make sure that that he hasn't grown too much for his seat. Otherwise, he could hurt or break his neck or his back when he lands.

Franks: Oh, it's that serious? I mean it's not just this is a first world problem about trying to fit in an economy class seat. This is something potentially rather more serious. And how quickly on return do you get back to normal size?

Hadfield: Well, you know, I actually when I landed the first time on my first space flight, as I started getting under gravity and I stood up again, I swore I could actually feel like when you slip something in your back. I could feel my back compressing back down again - almost a palpable give to it. And it happens quite rapidly. Gravity is relentless. It's a punishing oppressor, and it grinds our bones as close to them as our body will allow, so no matter how much taller you get in space, it's not going to last when you get home.

Presenter: That's the former Canadian astronaut Chris Hadfield speaking to the BBC's Tim Franks.

Answer Key

Comprehension Activity

1. Three weeks
2. 9 centimeters
3. b) There is no gravity to stand straight
4. b) Their backbone stretches without gravity
5. About 4 centimeters
6. b) It affects the astronaut's safety during landing
7. The neck or back
8. c) Quite rapidly
9. b) A punishing oppressor

Reading Comprehension

1. Because there is no gravity, so the body doesn't need to support weight.
2. It stretches and the vertebrae separate
3. Cancer.

Idioms

“More than reality” = Not completely accurate or exaggerated

“Protective atmosphere” = The layer of gases that guards Earth from harmful space conditions